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IMPROVEMENT IN SELECTION PROCESS BASED ON TRANSLATION EXAMPLES OF GA-ILMT

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ABSTRACT

The goal of our research is to realize a machine translation system that is capable of adapting itself to the environment. For this purpose, we proposed a method of machine translation using inductive learning with genetic algorithms, and we evaluated the method. As the results of experiments, we confirmed the effectiveness of applying genetic algorithms. However, the system based on this method produces many erroneous translation rules that cannot be completely removed from the dictionary. To resolve this problem, we improve the selection process based on translation examples. In this paper, we describe this improvement in the selection process and the results of evaluation experiments.

Key Words: machine translation, inductive learning, genetic algorithms, selection process, translation examples

1 INTRODUCTION

Recently, many studies have been carried out on machine translation. However, there are several problems in the machine translation. Rule-based machine translation [1] could not deal adequately with various linguistic phenomena due to the use of limited rules. To resolve this problem, Example-based machine translation [2, 3, 4, 5] has recently been proposed. However, this method requires many translation examples to achieve a practical and high-quality translation.

The goal of our research is to realize a system that is capable of adapting itself to the environment [6, 7]. Towards this goal, we previously proposed a method of Machine Translation using Inductive Learning with Genetic Algorithms (GA-ILMT), and evaluated it [8]. In this method, translation rules are inductively acquired and trnslation results are produced by using these rules. Many new translation examples are automatically produced from only a small number of translation examples by applying genetic algorithms to the method of machine translation using inductive learning. We consider that our proposed method can solve problems that Rule-based machine translation cannot deal with various linguistic phenomena and Example-based machine translation would require many translation examples.

However, the results of the evaluation experiments show that this method has some problems. The main problem is that many erroneous translation rules are produced and these rules cannot be completely removed from the dictionary. This leads to many erroneous translation results. The main reason for this is that the capability of the selection process is not sufficient to remove those erroneous translation rules. Therefore, we need to improve the selection process of GA-ILMT. In this paper, we describe an improvement in the selection process of GA-ILMT based on translation examples and the effectiveness of the improvement.

2 OUTLINE OF TRANSLATION METHOD

2.1 OVERVIEW

Figure 1 shows the outline of our proposed translation method. First, the user inputs a source sentence in English. Second, in the translation process, the system produces several candidates of translation results using translation rules extracted in the learning process. Third, the user proofreads the translated sentences if they include some errors. Fourth, in the feedback process, the system determines the fitness value of translation rules used in the translation process and performs the selection process of erroneous translation rules. In the learning process, new translation examples are automatically produced by crossover and mutation, and various translation rules are extracted from the translation examples by inductive learning. There are two kinds of translation rules: those for sentences and those for words. The former are called sentence translation rules and the latter word translation rules. Repetition of the above-mentioned processes corresponds to generational replacement of the whole system, and the system continuously evolves to higher quality translation.

2.2 CHROMOSOME AND GENES

As shown in Figure 2, a chromosome corresponds to a translation example that consists of an English and a Japanese sentence, and a gene corresponds to a word. In this paper, Japanese words are written in italics.

3 IMPROVEMENT IN SELECTION PROCESS

3.1 PROBLEM

The selection process is performed in the feedback process described in Section 2.1. First, the system evaluates the translation results using the translated sentences which have been proofread. The system adds one to the correct translation frequency when the translation results have the same character strings as the proofread translation results, and adds one to the erroneous translation frequency when the translation results have different character strings to the proofread translation results. Second, the system determines the fitness value of the rules used in translation using these correct and erroneous translation frequencies.

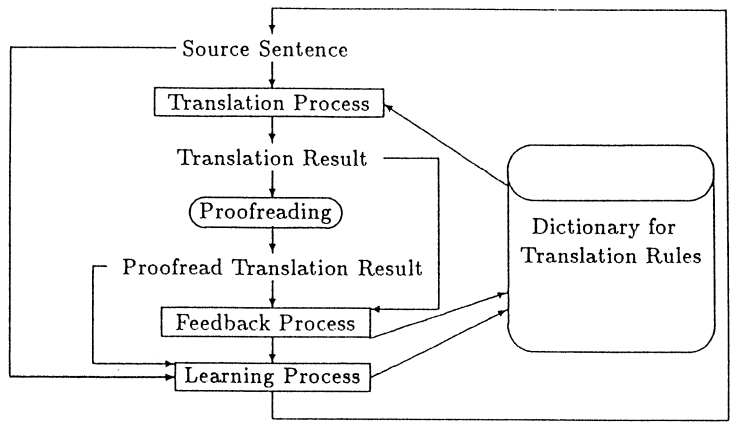


Figure 1: OUTLINE OF THE TRANSLATION METHOD

The fitness value is calculated by the fitness function as follows:

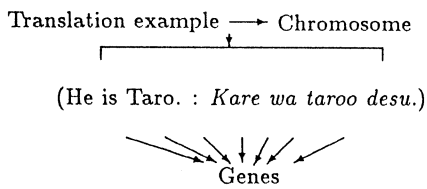


Figure 2: CHROMOSOME AND GENES

$$\text{Fitness value}(\%) = \frac{\text{The correct translation frequency}}{\text{The number of uses}} \times 100 \quad (1)$$

Third, the system performs the selection process using the fitness value.

The previous selection process have some problems. First, the translation rules are evaluated only when they are used in the translation process, and they are a few translation rules of all the translation rules in the dictionary. Therefore, the removed erroneous translation rules are a few erroneous translation rules of all the erroneous translation rules in the dictionary. Second, most of removed erroneous translation rules consist of a few words. Most of the erroneous translation rules which consist of many words have different character strings or structure to the source sentence, and cannot be selected as the translation rules which can apply to the source sentence in the translation process. Therefore, the system based on the previous selection process is difficult in removing the erroneous translation rules which consist of many words from the dictionary.

3.2 IMPROVEMENT

To resolve this problem described in Section 3.1, we propose an improvement for the selection proces of GA-ILMT. The execution of the selection process in genetic algorithms [9, 10] depends on whether an individual adapts to the environment or not. The individual which does not adapt to the environment is removed in the selection process. In GA-ILMT, the given translation examples correspond to the environments. Therefore, our proposed method utilizes given translation examples. Also, this method does not require any analytical knowledge as initial condition, such as an English-Japanese dictionary. The reason is that the method which uses analytical knowledge has the problem, such as difficulty in dealing with unregistered words. We consider that our proposed method can remove most of erroneous translation rules by learning method which utilizes only given translation examples.

Our proposed improvement evaluates the translation rules by utilizing the given translation examples directly. Namely, it determines whether a combination of the English word and the Japanese word in a translation rule is true or false by utilizing the given translation examples. Figure 3 shows examples of the "true" and the "false" combinations of words.

In Figure 3, the combination of words which are "I" in English and "*watashi* (In Japanese "I")" in Japanese may be true when it exists in a given translation example. On the other hand, the combination of words which are "volleyball" in English and "*basukettobooru*(In Japanese "basketball")" in Japanese is false when it does not exist in all given translation examples. In all combinations of words in a translation rule, the system determines whether the each combination of words is true or false. The system determines the rate of error based on the number of "false" combinations of words, and removes the translation rules for which the rate of error is high. The details of this process are as follows:

1. The system produces all combinations of words between the English words and Japanese words in a translation rule.

2. The system determines whether the combinations of words exist in given translation examples or not.

3. The system gives "true" for the combinations of words when they exist in a given translation example, and gives "false" for the combinations of words when they do not exist in all given translation examples.

In these "false" combinations, there are two possibilities for faults as follows : (1) an erroneous word used in translation (2) an erroneous combination. For these faults, the system performs the process 4.

4. The system determines the combinations of words which have the erroneous combinations and deletes them. The details of this process are as follows:

(a) The system changes the combinations of words between the English words and the Japanese words in all "false" combinations.

(b) The system searches the changed combinations of words in translation rules in the dictionary.

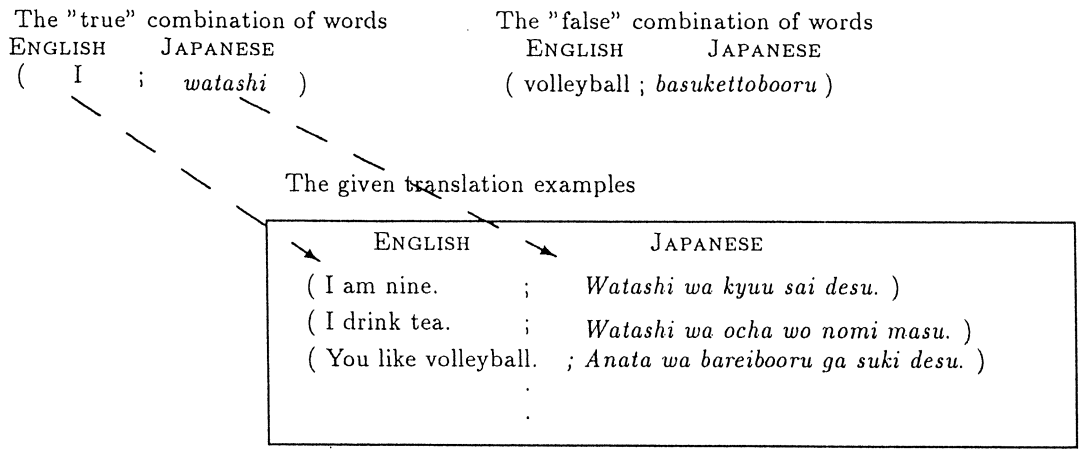


Figure 3: EXAMPLES OF THE "TRUE" AND THE "FALSE" COMBINATIONS OF WORDS

Table 1: EXAMPLE OF THE ERRONEOUS TRANSLATION RULE (SHE IS NOT MY TEAMMATE.; *KANOJYO WA BOKU NO CHIIMUMEITO DESU*[IN JAPANESE "SHE IS MY TEAMMATE."].)

	she	is	not	my	teammate
kanojyo	true	true	true	true	true
wa	true	true	true	true	true
boku no	true	true	true	true	true
chiimu-meito	true	true	true	true	true
desu	true	true	false	true	true

(c) The system changes "false" for "true" in the results of evaluations for the "false" combinations of words when the changed combinations of words exist in the translation rules in the dictionary. Table 1 shows an example of the erroneous translation rule.

5. The system determines the rate of error based on the number of the "false" combinations of words. The rate of error is calculated as follows:

$$\text{Rate of error}(\%) = \frac{\text{The number of "false" combinations}}{\text{The number of combinations}} \times 100 \qquad (2)$$

In Table 1, the rate of error is 4.0%(1/25×100).

6. The system determines the translation rules for which the rate of error is over the threshold, and removes them from the dictionary. The condition of removal is defined as follows:

$$\text{Rate of error} \geq x \qquad (x : \text{Threshold}) \qquad (3)$$

4 EXPERIMENTS

We evaluate the effectiveness of our proposed improvement for removing the erroneous translation rules in the selection process of GA-ILMT, and the performance of GA-ILMT using translation examples.

Table 2: PRECISION AND RECALL WHICH WERE USED SOME THRESHOLDS

	Precision	Recall
Threshold is 2%	92.6%	66.4%
Threshold is 5%	94.5%	61.3%
Threshold is 10%	96.7%	52.0%
Threshold is 15%	97.6%	40.5%

4.1 METHOD OF EXPERIMENTS

In the experiments, 550 translation examples were used as data. Of these, 500 were used to decide the best threshold. The examples were taken from a textbook [11] for first-grade junior high school students. All of the translation examples were processed by the method outlined in Figure 1. The initial dictionary was empty. The experiments were carried out with and without the improvement for the selection process described in Section 3.2.

4.2 RESULTS OF EXPERIMENTS

Table 2 shows the precision and the recall which were gotten by using some thresholds. The precision is the percentage of removed erroneous translation rules to removed translation rules, and the recall is the percentage of removed erroneous translation rules to erroneous translation rules in the dictionary. Table 3 shows the precision and the recall with and without the improvement for the selection process, and Table 4 shows the rate of translation with and without the improvement for the selection process. In Table 3 and Table 4, 5% is used as the threshold in the results of the method with the improvement for the selection process.

5 DISCUSSION

In Table 2, the precision increases by about 2%. On the other hand, the recall decreases by about 10% when the threshold is over 5%. However, the recall decreases by 5.1% when the threshold is under 5%. Therefore, we decided that the best threshold was 5% because the rate of the decrease in the recall is lower when the threshold is under 5% than over 5%.

Table 3: PRECISION AND RECALL WITH AND WITHOUT THE IMPROVEMENT FOR THE SELECTION PROCESS

	Precision	Recall
Method without improvement	85.6%	4.6%
Method with improvement	95.7%	62.8%

Table 4: RATE OF TRANSLATION

	Method without improvement	Method with improvement
Effective transaltion	48.8%	50.8%
Ineffective translation	51.2%	49.2%

In Table 3, the precision increases from 85.6% to 95.7% and the recall increases from 4.6% to 62.8%. These results show that our proposed improvement could resolve the problem that the removed erroneous translation rules are a few erroneous translation rules of all the erroneous translation rules in the dictionary. In the previous selection process, the translation rules are evaluated only when they are used in the translation process. Therefore, the translation rules which are not used in the translation process are never removed from the dictionary. However, the proposed improvement can evaluate all of the produced translation rules by utilizing only the given translation examples. In the previous selection process, the rate of the evaluated translation rules is 1.7% of the translation rules in the dictionary.

Also, our proposed improvement could resolve the problem that most of the erroneous translation rules which consist of many words cannot be removed from the dictionary. In the previous selection process, the average number of words in a removed translation rule is 4.3. However, the average number of words in a removed translation rule is 7.6 in our proposed improvement. The average number of words in a translation rule in the dictionary is 7.4. Therefore, we consider that our proposed improvement is effective for the erroneous translation rules which consist of many words.

In Table 4, the rate of translation increases from 48.8% to 50.8%. In GA-ILMT, the system ranks from the first to the tenth of the candidates for the translation results. The effective translation results are included in these ranks. In the effective translation, the number of the translation results that the rank was under the first was 117. Of these, 87 improved their ranks in the translation results. Therefore, the rate of the translation results that the rank improved was 74.4%. The reason for this is that the system based on our proposed improvement can remove many erroneous translation rules in the dictionary.

6 CONCLUSION

In this paper, we proposed an improvement in the selection process of GA-ILMT. The results of evaluation experiments showed that the precision increases from 85.6 to 95.7% and the recall increases from 4.6% to 62.8%. These results show that our proposed improvement can remove many erroneous translation rules without the requirement of any analytical knowledge as initial condition. Therefore, we confirmed that our proposed improvement is effective to improve the selection process of GA-ILMT. In the future, we plan to experiment using the various data, and improve GA-ILMT to get the high correct rate and quality of translation.

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